# Vanshika Srivastava

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## Education

**Shiv Nadar University, CGPA: 8.00** — Greater Noida, India
*Bachelor of Technology in Computer Science, Minor in Management* — *Aug. 2023 – May 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Software Testing, Machine Learning, Information Retreival

## Experience

**Software Testing Intern** — Apr 2026 – May 2026
*Buddy4Travel, Australia* — *Remote*

- Executed manual, load and volume testing with **JMeter** and **Postman** across app releases, surfacing **40+ defects** before launch and triaging them by severity so critical issues were fixed first.
- Served as QA point of contact for a remote Australian client team, running UAT verification on **TestFlight** builds across **10+ release cycles** .

**Project Intern** — May 2025 – Jul 2025
*Defence Research and Development Organisation (DRDO)* | GitHub — *Dehradun*

- Built a real-time network intrusion detection system for defence networks: **Scapy** captures live traffic, rebuilds flows and extracts 25 features, each finished flow is classified and logged to **SQLite**, and a **Flask** backend feeds a dashboard that refreshes every 5 seconds.
- Engineered a memory-efficient data pipeline with chunked **Pandas** loading and feature selection (**25 of 80+** features) to process **2M+** network flows within **8GB RAM**, rebalancing rare attack classes with **SMOTE**.
- Trained a soft-voting **XGBoost**, **LightGBM** and **Random Forest** ensemble reaching **99.85% weighted F1** (0.926 macro) on held-out data.

**Undergraduate Teaching Assistant, Artificial Intelligence** — Aug 2026 – Present
*Shiv Nadar University* — *Greater Noida*

- Guide **50+** students through weekly doubt-clearing sessions and write assignments on **search algorithms** and **supervised learning**.

## Projects

**PropVisit: Property Visit Booking Platform** | *Python, FastAPI, PostgreSQL, Redis, React, TypeScript, Docker* | GitHub — Jul. 2026

- Developed a full-stack visit-booking platform with **24** REST and WebSocket endpoints, where buyers find listings on a **Leaflet** map and reserve viewing slots, using **FastAPI**, **PostgreSQL** and **React** + **TypeScript**.
- Prevented double-booking by pairing a 5-minute **Redis** slot hold with a partial unique index in **PostgreSQL**, then proved it with concurrent-request tests and by deliberately breaking six safeguards to confirm the tests caught each one.
- Added Redis search caching, live agent alerts over **WebSockets** and a queued email worker, backed by **221** tests (**pytest**, **Vitest**, **Playwright**) in **GitHub Actions** CI, with backend tests running against real PostgreSQL and Redis.

**AI Claims Investigator Copilot: Insurance Fraud Analytics** | *Python, MCP, XGBoost, SHAP, SQL, MySQL* | GitHub — Aug. 2026 – Oct. 2026

- Architected an **MCP server** exposing four tools: schema, AST-guarded **SQL**, **SHAP**-explained fraud scoring, and hybrid **BM25** + vector search over IRDAI and California fraud guidelines.
- Verified with **47** tests, **19** SQL-injection and 20-question retrieval eval (**0.95** hit@5), and caught dataset artifact inflating ROC-AUC from 0.76 to 0.86.

**SONIC: Epidemic Containment via Source Inference** | *Python, PyTorch Geometric, NetworkX, SciPy* | GitHub | Demo — Apr. 2026

- Designed **SONIC**, source-aware epidemic containment combining **DeepTrace** GNN source detection, **Expected Personalized PageRank** risk scoring and spectral-radius node removal; built the **SIS** simulator and experiments.
- Outperformed **DINO** by **11%** and **Degree** by **25%** in spectral-radius reduction on the real-world **Gnutella** network (10.9K nodes), beating seven baselines at 200 immunized nodes.

**The Route Cause: AI Traffic Signal Management** | *Python, YOLOv8, Q-Learning, FastAPI, React, TypeScript* | GitHub — Sep 2025

- Co-developed an AI traffic-signal control system: **YOLOv8** and **OpenCV** count vehicles and pedestrians in traffic video, feeding a **Q-learning** agent that selects signal phases from lane densities, current phase and elapsed time across 12,500 discrete intersection states.
- Paired the agent with a rule-based safety layer for emergency preemption and starvation prevention, streaming live decisions over **FastAPI** WebSockets to a **React** and **TypeScript** dashboard.

**Content-Aware Caching Algorithm** | *Java, Algorithms, Systems* | GitHub — Apr. 2025

- Devised a **Java** content-aware file cache that evicts by a weighted score of file type, size, access frequency and recency, with configurable per-type priorities, raising hit rate from **46.4%** to **74.5%** versus LRU over 20,000 simulated accesses (**78.3%** to **93.2%** on important-file bursts).

## Accomplishments

- Ranked among the **top 20 teams out of 1,700+** in BrAINWARS, a competitive case study competition hosted by Bain Capability Network.
- Engineered a real-time hospital bed availability web app, securing a **top 3 finish at Hackdata**, an internal hackathon at SNU.
- Solved **300+ problems** on LeetCode and other platforms; selected participant at McKinsey Forward Programme 2026; Technical Team member ACMW.

## Certifications

- Oracle Certified Foundations Associate: **Agentic AI** ↗
- Oracle Certified Foundations Associate: **AI Database** ↗

## Technical Skills

**Languages & Frontend:** Python, Java, TypeScript, JavaScript, SQL, HTML/CSS, React, Vite, Tailwind CSS, Leaflet, Recharts
**Data Science & ML:** Pandas, NumPy, SciPy, Scikit-learn, EDA, data cleaning & preprocessing, feature engineering, hyperparameter tuning, Matplotlib
**BackendSystem Design:** FastAPI, Flask, REST APIs, WebSockets, JWT authentication, RBAC, caching strategies, ACID transactions, rate limiting, API design
**Databases:** PostgreSQL, MySQL, SQLite, Redis, SQLAlchemy, database schema design, indexing
**AI, ML & GenAI:** PyTorch, PyTorch Geometric (GNNs), reinforcement learning (Q-learning), computer vision (YOLOv8, OpenCV), RAG, MCP, hybrid search (BM25 + vectors), TF-IDF ranking, prompt engineering
**Tools & Platforms:**Git, GitHub Actions, Docker, Supabase, Streamlit, pytest, JUnit, Selenium, Jest, Postman, JMeter , UAT, TestFlight, Agile, Claude Code, SDLC